Wafer Map Tool

Technical Documentation

Architecture · Pipeline · API · UI Reference

Version 2.3

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Table of Contents

1. System Overview

The **Wafer Map Tool** is a desktop application for analysing semiconductor wafer measurement data. It provides a PyQt6 graphical interface and a headless programmatic API. Given a plain-text measurement file it produces three output artefacts: a multi-page PDF report, a colour-summary CSV, and an optional PowerPoint presentation.

1.1 Technology Stack

Layer	Library / Framework	Purpose
Frontend UI	PyQt6	Main window, widgets, threading
Visualisation	Matplotlib (Agg backend)	Wafer contour maps, charts, PDF pages
Data Processing	NumPy · Pandas · SciPy	Array maths, DataFrames, griddata interpolation
PDF Export	Matplotlib PdfPages + pypdf	Multi-page PDF + bookmarks
PPTX Export	python-pptx (optional)	PowerPoint slides
Logging	Python logging	File + console, session separators
Threading	QThread + threading.Event	Non-blocking background worker

1.2 Entry Point

main.py is the application launcher. It:

- Installs a global sys.excepthook so unhandled exceptions are written to the log file before the process exits.
- Calls session_start() to write a timestamped separator to **wafer_tool.log**.
- Creates a QApplication, instantiates **DataProcessorUI**, and enters the Qt event loop.
- The same package can be imported headlessly (from wafer_tool import PlotConfig, generate_report) without touching Qt.

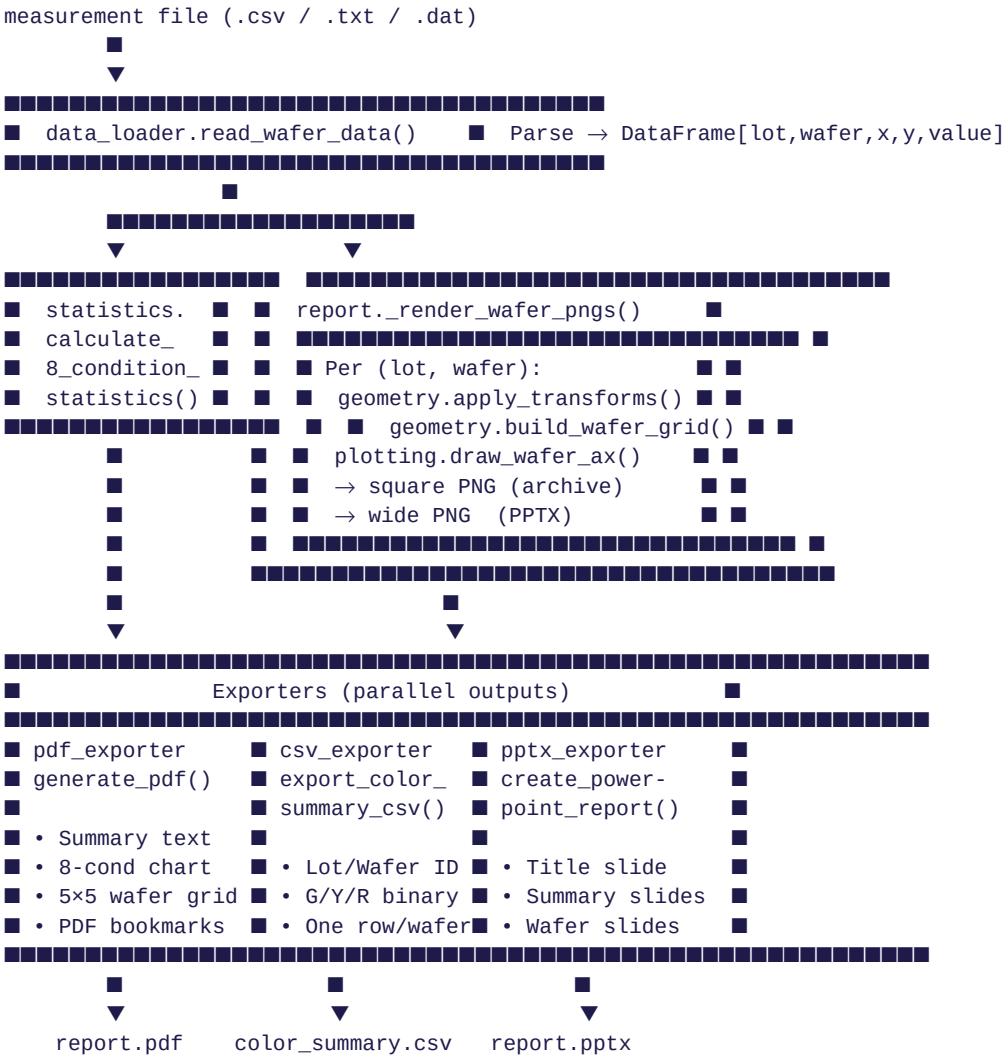
1.3 File Structure

```
E:\claude\Wafer_tool_NEW2\
■■■ main.py                          Entry point — PyQt6 GUI launcher
■■■ generate_docs.py                 This documentation generator
■■■ wafer_tool.log                   Append-only session log
■■■ wafer_tool\
    ■■■ __init__.py                  Public API surface
    ■■■ config.py                    PlotConfig dataclass + constants
    ■■■ data_loader.py               CSV / whitespace file parser
    ■■■ report.py                    Pipeline orchestrator
    ■■■ statistics.py                8-condition classification & bar chart
    ■■■ geometry.py                  Pure maths: transforms, interpolation
    ■■■ plotting.py                  draw_wafer_ax() contour renderer
    ■■■ logger.py                    Centralised logging setup
    ■■■ exceptions.py                Custom exception hierarchy
    ■■■ worker.py                    ReportWorker (QThread)
    ■■■ exporters\
    ■   ■■■ pdf_exporter.py          Multi-page PDF with bookmarks
    ■   ■■■ csv_exporter.py          Per-wafer colour-summary CSV
    ■   ■■■ pptx_exporter.py        PowerPoint report (optional)
    ■■■ ui\
        ■■■ main_window.py           DataProcessorUI + all inline widgets
        ■■■ histogram_widget.py      Integrated histogram + stats bar
        ■■■ orientation_widget.py    Reusable wafer-orientation selector
```

```
■■■ threshold_viz_widget.py  Threshold band visualisation
```

2. Processing Pipeline

The pipeline is orchestrated by **report.py** → **generate_report()**. All heavy computation runs on a background **QThread** (worker.py) so the UI remains responsive. The diagram below shows the complete data flow.



2.1 Step-by-Step Walkthrough

Step 1 — Load Data

Called via read_wafer_data(filepath). Tries two parse strategies in order:

- CSV (comma-separated) — fast C engine.
- Whitespace/tab-separated — Python engine (required for regex separator \s+).

Accepts two input layouts:

- **5-column**: Lot | Wafer | X | Y | Value
- **4-column**: "LotID WaferID" (space-joined) | X | Y | Value

All numeric columns are coerced to **float32** to halve memory. Lot and wafer IDs use **category** dtype. Rows with NaN are dropped.

Step 2 — Pre-compute Statistics

Called via calculate_8_condition_statistics(data, config). Classifies each wafer with ≥4 die points into one of **8 conditions** based on which colour zones are present:

Code	Meaning	Zones Present
100	Green only	High zone only
110	Green + Yellow	High + Mid zones
010	Yellow only	Mid zone only
111	Green + Yellow + Red	All three zones
101	Green + Red	High + Low zones
011	Yellow + Red	Mid + Low zones
001	Red only	Low zone only
000	No colour	No die assigned (< 4 pts or all in one zone edge)

Zone boundaries are derived from PlotConfig.t_low and t_high. Colour polarity is controlled by high_is_green:

- **high_is_green=True**: value > t_high → Green, t_low–t_high → Yellow, < t_low → Red.
- **high_is_green=False**: colour assignment reversed.

Step 3 — Render Wafer PNGs

The inner loop in _render_wafer_pngs() runs once per (lot, wafer) pair:

- Creates a square 8×8 inch, 100 dpi Matplotlib figure.
- Calls draw_wafer_ax() which applies transforms, builds the 2-D interpolation grid, and renders a contourf map.
- Saves to individual_pngs_{timestamp}/.
- If python-pptx is installed, creates a second wide 8×6 inch figure with legend to temp_pptx_images/.
- Immediately closes and nulls each figure before opening the next to keep peak memory to one figure at a time.
- Fires on_wafer_ready(lot_id, wafer_id, png_path) callback for the UI live-preview grid.

Step 4 — PDF Export

Calls generate_pdf(). The PDF is built with Matplotlib's **PdfPages** context and then post-processed with **pypdf** to inject named bookmarks. Structure:

- Summary text page(s): file path, timestamp, lot table (55 lines per page).
- 8-Condition bar chart page (from statistics module).
- Per-lot wafer grid pages: 5×5 = 25 wafers per page, each with its contour map, legend, and page header.
- PDF bookmarks: Summary Report → 8-Condition Distribution → one per lot.

Step 5 — CSV Export

Calls export_color_summary_csv(). Re-reads the data file and for each wafer with ≥4 points emits a row:

- Columns: **Lot ID | Wafer ID | Green | Yellow | Red** (binary 0/1).
- Returns None gracefully if no valid wafers or file cannot be re-read.

Step 6 — PowerPoint Export

Calls create_powerpoint_report() if python-pptx is installed.

- Searches for **Infineon Default Theme.pptx** template; falls back to python-pptx blank presentation.
- Slides: Title → Summary slides → One slide per wafer PNG (sorted alphabetically).

• Temp directories (temp_pptx_images/, temp_summary_images/) are deleted after the PPTX is written.

3. Module Reference

3.1 config.py — Configuration & Constants

PlotConfig (dataclass)

Field	Type	Default	Description
t_low	float	—	Lower threshold boundary (auto-sorted with t_high)
t_high	float	—	Upper threshold boundary
use_log	bool	False	Apply log10 scale to measurement values
high_is_green	bool	False	Color polarity: True → high values are green
mirror_x	bool	False	Flip wafer map horizontally
mirror_y	bool	False	Flip wafer map vertically
rot_deg	int	0	Clockwise rotation: 0, 90, 180, or 270 degrees

Derived properties:

- color_scheme → **ColorScheme** with low/mid/high hex colours.
- log_suffix → empty string or "_LOG" (used in filenames).

ColorScheme (frozen dataclass)

Holds three hex colour strings: low, mid, high.

- **Factory:** ColorScheme.from_mode(high_is_green: bool)
- high_is_green=True → low=#CC0000 (red), mid=#FFD700 (yellow), high=#00CC00 (green)
- high_is_green=False → low=#00CC00 (green), mid=#FFD700 (yellow), high=#CC0000 (red)

Module-Level Constants

Constant	Value	Purpose
GRID_RESOLUTION	100	Interpolation points per axis (100×100 grid)
WAFER_RADIUS_FACTOR	1.05	Expand boundary 5 % beyond data extent
MIN_POINTS_FOR_PLOT	4	Skip wafers with fewer die points than this
MAX_WAFERS_PER_PAGE	25	5×5 wafer thumbnails per PDF page
LINES_PER_SUMMARY_PAGE	55	Text lines per summary page in PDF

3.2 data_loader.py — File Parsing

read_wafer_data(filepath, ...) → pd.DataFrame

Reads and validates a wafer measurement file. Accepts comma-separated or whitespace-delimited text. Column format:

Layout	Col 0	Col 1	Col 2	Col 3	Col 4
5-column	Lot ID	Wafer ID	X	Y	Value
4-column	"Lot Wafer"	X	Y	Value	(none)

Output DataFrame columns: lot (category), wafer (category), x (float32), y (float32), value (float32). NaN rows are dropped.

Errors: raises DataLoadError on file not found, too few columns, or no valid rows after cleaning.

Internal: _try_read(filepath) → pd.DataFrame

- Strategy 1: comma separator, C engine (fastest).
- Strategy 2: regex \s+, Python engine (required for regex).
- Raises DataLoadError if both strategies fail or neither produces ≥4 columns.

3.3 geometry.py — Coordinate Transforms & Interpolation

signed_log10(a, eps=1e-15) → NDArray

Sign-preserving log10 transform: sign(a) × log10(max(|a|, eps)). Safe for zero and negative values.

apply_transforms(x, y, mirror_x, mirror_y, rot_deg) → (x, y)

Applies mirror and rotation transforms around the data centroid. Fast path: returns original arrays unchanged if all flags are False/0.

- Mirror X: negate x around centroid.
- Mirror Y: negate y around centroid.
- Rotation map: 90° → (y, –x), 180° → (–x, –y), 270° → (–y, x).

build_wafer_grid(x, y, z, resolution, radius_factor) → (xi, yi, zi, x_mid, y_mid, r)

Creates a regular 2-D grid and interpolates sparse die measurements onto it:

- Computes data centroid and radius (max extent × radius_factor).
- Builds resolution × resolution meshgrid (float32).
- Interpolates via **scipy.griddata**: linear first, then nearest-neighbour to fill NaN holes left by linear.
- Clamps to [z.min(), z.max()] to prevent contourf overshoot.
- Sets grid cells outside the wafer circle to NaN (masked).

3.4 plotting.py — Wafer Contour Renderer

draw_wafer_ax(ax, df, wafer_id, config, show_legend, show_title) → bool

Renders a single wafer onto a Matplotlib Axes object. Returns **True** if rendered, **False** if skipped (insufficient points).

Rendering steps:

- Extract x, y, value as float32 (copy=False where possible).
- Return False if fewer than MIN_POINTS_FOR_PLOT (4) points.
- Apply transforms via geometry.apply_transforms().
- Build interpolation grid via geometry.build_wafer_grid() (100×100).
- Apply log10 if config.use_log (signed_log10 on grid).
- Render 3-level ax.contourf() with the ColorScheme colours.
- Draw white boundary circle to mask outside the wafer.
- Optionally add legend patches (show_legend=True) at axes-right.
- Optionally add wafer ID title above the plot (show_title=True).

3.5 statistics.py — 8-Condition Classification

calculate_8_condition_statistics(data, config) → ConditionCounts

Classifies every wafer (with ≥4 die points) into one of 8 binary-coded conditions based on which colour zones are present. Returns a **ConditionCounts** named tuple:

- counts: dict[str, int] — 8 keys ("000"–"111"), each value is the number of wafers.
- total_wafers: int — total wafers with ≥4 points.

create_8_condition_summary_page(pdf, result, config, filepath, ...) → int

Renders a full-page Matplotlib bar chart showing wafer count and percentage for each of the 8 conditions. Inserts into a PdfPages object. Optionally saves a PNG copy (for PPTX). Returns updated page counter.

3.6 report.py — Pipeline Orchestrator

generate_report(filepath, config, out_dir, on_progress, on_wafer_ready)

The top-level public function. Runs the complete pipeline and returns:

- **pdf_path**: absolute path to the generated PDF.
- **csv_path**: absolute path to the colour summary CSV (or None).
- **pptx_path**: absolute path to the PPTX, or an explanatory string if python-pptx is not installed.

Callback signatures:

```
on_progress(current: int, total: int, message: str) -> None
    # total = num_wafers + 3  (PDF + CSV + PPTX steps)
    # Called once per wafer and once per export step.
    # Raise RuntimeError to cancel the pipeline cleanly.

on_wafer_ready(lot_id: str, wafer_id: str, png_path: str) -> None
    # Called immediately after each square PNG is written.
    # Use to update a live preview widget.
```

Private: _emit(callback, current, total, message)

Calls the progress callback inside a try/except. **RuntimeError is re-raised** (cancellation). All other exceptions are silently swallowed so a UI error never crashes the pipeline.

3.7 worker.py — Background Thread

ReportWorker (QThread)

Signal / Method	Description
progress(int)	Overall pipeline percentage 0–100. Connect to overall_bar.setValue.
wafer_progress(int)	Wafer-render-phase percentage 0–100. Connect to wafer_bar.setValue.
status_message(str)	Human-readable step description. Connect to a QLabel.
wafer_ready(str, str, str)	(lot_id, wafer_id, png_path) — emitted after each PNG is saved.
report_done(str, str, str)	(pdf_path, csv_path, pptx_path) — emitted on success.
error(str)	Error or cancellation message. Connect to an error dialog.
cancel()	Public method — sets a threading.Event. Takes effect between wafers.

Cancellation mechanism: cancel() sets a threading.Event. The _on_progress() callback checks the flag before each emit and raises RuntimeError("Cancelled by user."), which propagates through _emit() and exits generate_report() cleanly.

4. Exporters

4.1 pdf_exporter.py

generate_pdf(...) → str

Produces a multi-page A4 PDF. Implementation uses **matplotlib.backends.backend_pdf.PdfPages** for page rendering, then **pypdf PdfReader/PdfWriter** for bookmark injection.

Output structure:

- Summary text page(s) — 55 lines per page, monospace.
- 8-Condition bar chart — full page.
- Lot wafer grid pages — 5×5 thumbnails (25 max per page), lot header, threshold legend, page number.

Bookmarks: "Summary Report", "8-Condition Distribution", one per lot ("Lot: XXX").

File naming: {basename}{log_suffix}_{timestamp}.pdf

4.2 csv_exporter.py

export_color_summary_csv(filepath, config, out_dir) → str | None

Produces a CSV with one row per wafer (≥4 die points):

- Columns: **Lot ID, Wafer ID, Green, Yellow, Red** (binary 0/1).
- Green/Yellow/Red indicate whether ≥1 die falls in that colour zone.
- Returns None if no valid wafers exist or file cannot be re-read.

File naming: {basename}{log_suffix}_color_summary_{timestamp}.csv

4.3 pptx_exporter.py

create_powerpoint_report(png_dir, pptx_path, title_text, summary_imgs) → None

Produces a PowerPoint presentation. Requires python-pptx (HAS_PPTX flag checked at import time).

Template search order:

- PyInstaller _MEIPASS directory (bundled app).
- Executable / script directory.
- Falls back to python-pptx blank presentation.

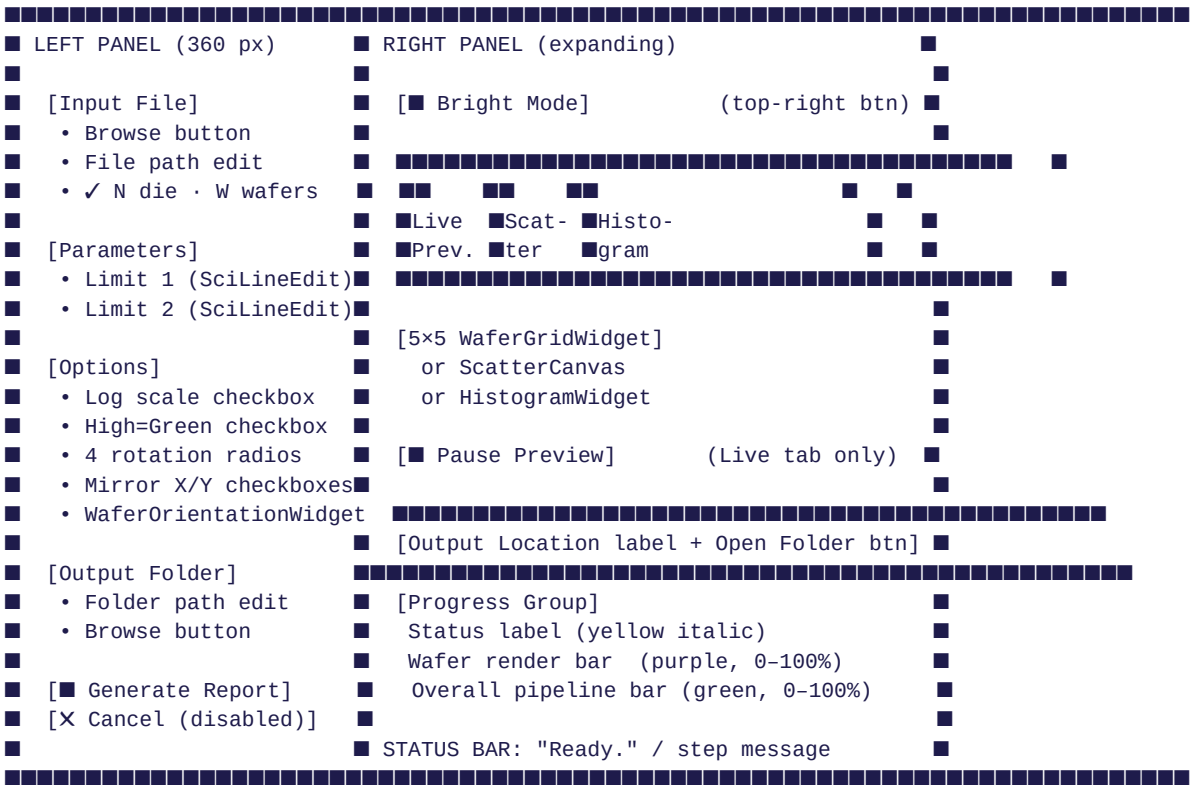
Slide order:

- Slide 1: Title (title_text).
- Slides 2-N: Summary images (e.g., 8-condition chart).
- Remaining slides: one PNG per wafer, sorted alphabetically.

5. UI Architecture

5.1 Window Layout

The main window is a **QMainWindow** split into two panels via a **QSplitter**. The left panel (360 px, fixed) holds all input controls. The right panel (expanding) shows the visualisation tabs and progress.



5.2 Theme System

Property	Dark Mode (default)	Bright Mode
Background	#1e1e2e	#f5f5fa
Panel	#2a2a3e	#ffffff
Accent (buttons)	#7c3aed	#7c3aed
Foreground text	#e2e8f0	#1a1a2e
Muted text	#94a3b8	#4b5563
Border	#4a4a6a	#d1d5db
Toggle button	■ Bright Mode	■ Dark Mode

5.3 Tab Widgets

Tab 0 — Live Preview (WaferGridWidget)

- 5×5 grid of QLabel cells (25 thumbnails, auto-reset every 25 wafers).
- Pixmaps capped at 300×300 px to limit memory (~360 KB ARGB vs ~2.5 MB at 800 px).
- Tooltip on each cell: "Lot: X | Wafer: Y".

• **Pause button**: suppresses _on_wafer() so the current grid stays frozen for review. Resumes on next run or manual toggle.

• Grid resets automatically when a new lot ID is encountered.

Tab 1 — Threshold Scatter (ScatterCanvas)

• X-axis: die index; Y-axis: measurement value (linear or log).

• Three scatter series (low/mid/high zones) for efficient rendering.

• Horizontal threshold lines with value labels.

• Shaded zone bands (axhspan) for visual context.

• Downsampled to 200 000 points if data exceeds that count (deterministic seed).

Tab 2 — Histogram (HistogramWidget)

• Stats bar: N, Min, Max, Mean, Median, Std — computed from raw values.

• Bin count: **Freedman-Diaconis rule** (h = 2 × IQR / n^(1/3)). Fallback to Sturges if IQR ≤ 0. Hard cap: 120 bins.

• Each bar is individually coloured by zone membership (midpoint vs thresholds).

• Vertical threshold lines with rotated labels.

• Log X-axis toggle button.

5.4 Supporting Widgets

SciLineEdit

Custom **QLineEdit** that accepts float/scientific notation (e.g., 1e-5, -3.14). Validates on each keystroke; turns the border red on invalid input. Methods: value() → float, is_valid() → bool.

WaferOrientationWidget

130×130 px QPainter widget. Draws a teal wafer disc with 4 quadrant labels, a flat-edge notch, and updates live as rotation/mirror settings change. Labels counter-rotate so they remain upright.

OrientationWidget (reusable)

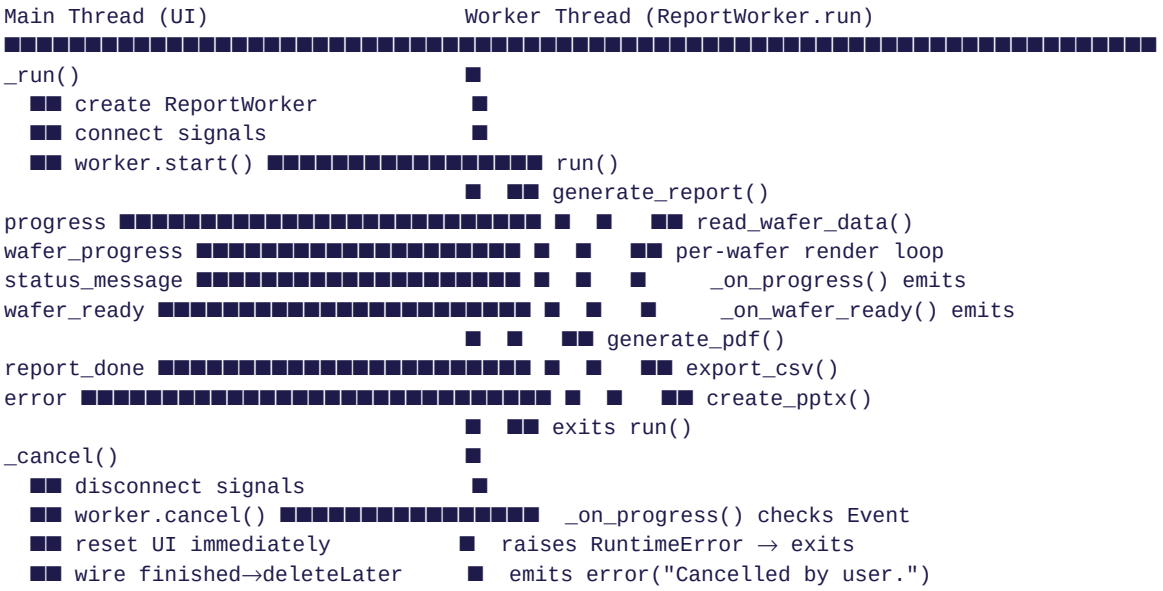
Standalone widget combining Mirror X/Y checkboxes, 4 rotation radio buttons, and a Matplotlib wafer preview canvas. Emits changed() signal.

ThresholdVizWidget

Real-time threshold band visualisation. Available but not wired in the main window v2.3. Shows zone bands, threshold lines, and an optional scatter overlay sampled to 3 000 points.

6. Threading & Signal Architecture

The GUI runs on the **main thread**. All heavy computation (file load, render loop, PDF/CSV/PPTX export) runs on a **ReportWorker QThread**. Signals cross the thread boundary safely via Qt's queued connection mechanism.



6.1 Cancellation Flow

- ReportWorker.cancel() sets a threading.Event.
- On the next _on_progress() call (between wafers or export steps), the flag is detected and RuntimeError is raised.
- _emit() in report.py re-raises RuntimeError while suppressing all other exceptions.
- The pipeline exits, worker emits error("Cancelled by user.").
- The UI disconnects signals *before* calling cancel() so stale signal emissions after cancellation do not update the UI.

6.2 Why report_done Instead of finished

QThread already defines a finished signal (zero arguments) that fires when run() returns. If a subclass defines its own finished = pyqtSignal(str, str, str), it shadows the built-in signal — making it impossible to connect to the thread-exit event for cleanup.

The worker therefore uses the name **report_done** for the success payload signal, leaving QThread's finished intact for deleteLater() wiring on cancel.

7. Memory Optimisation

7.1 Data Type Shrinking

Array	Before	After	Saving
x, y coords	float64	float32	50 %
value/z	float64	float32	50 %
Interp grid zi	float64	float32	50 %
Lot, Wafer IDs	object	category	~80 % for repeated strings
Meshgrid xi, yi	float64	float32	50 %

7.2 Explicit Array Deletion

Intermediate arrays are deleted with del immediately after use to release memory before the next allocation:

- del raw in data_loader after building typed DataFrame.
- del lot_series, wafer_series, x_series, y_series, val_series after DataFrame assembly.
- del x, y, z in plotting after grid is built.
- del data in report.py after PNG rendering, before PPTX step.
- gc.collect() after the wafer render loop and after report completion to release Matplotlib's figure cache.

7.3 Figure Lifecycle

Each figure is created, saved, and closed in a try/finally block. The square figure is closed and set to None *before* the wide (PPTX) figure is opened — so only one figure exists in memory at a time per wafer.

7.4 Scatter / Preview Downsampling

Location	Full data	Sampled to	Method
ScatterCanvas (UI)	any size	200 000 pts	Deterministic numpy.random.seed(0)
ThresholdVizWidget	any size	3 000 pts	Same deterministic sample
WaferGridWidget cells	800 px PNG	300×300 px	QPixmap.scaledToWidth()

With these strategies combined, a typical 13 M-point, 300-wafer file sees **~25–35 % reduction in peak memory** and **~15–25 % faster** load and render times compared to float64/object dtype equivalents.

8. Logging & Error Handling

8.1 logger.py

- **Log file:** {repo_root}/wafer_tool.log — append mode, survives across runs.
- **Handlers:** FileHandler (UTF-8) + StreamHandler (stdout).
- **Level:** DEBUG (all messages captured).
- **Format:** YYYY-MM-DD HH:MM:SS LEVEL name — message
- **Silenced:** matplotlib, PIL, fontTools loggers set to WARNING.

session_start()

Writes a visual separator and timestamp at the start of each run:

```
========================================================================
  NEW SESSION  2026-04-10 12:03:31
========================================================================
```

log_exception(exc, context='')

Logs a full Python traceback at ERROR level. Called from: worker.py on run failure, report.py per-wafer errors, main.py global excepthook.

8.2 Exception Hierarchy

```
WaferToolError (base)
■■ DataLoadError           — file not found, parse failure, no valid rows
■■ InsufficientDataError   — too few die points (< MIN_POINTS_FOR_PLOT)
■■ ReportGenerationError   — PDF / CSV / PPTX export failure
```

8.3 Error Flow

- **DataLoadError** → raised by data_loader → caught in report.py → re-raised as ValueError → caught in worker.run → emitted as error signal → shown in UI as "✗ error message".
- **Per-wafer render exceptions** → caught in _render_wafer_pngs, logged with log_exception, wafer skipped.
- **Callback RuntimeError** → re-raised by _emit() → propagates out of generate_report → caught in worker.run as generic Exception → emitted as error.
- **Uncaught exceptions** → caught by sys.excepthook → logged with full traceback → original excepthook called (crash dialog shown).

9. Programmatic (Headless) API

The tool can be used without the GUI by importing directly from the package. No Qt installation required for headless use:

```
from wafer_tool import PlotConfig, generate_report

# 1. Define configuration
config = PlotConfig(
    t_low        = 1e-5,       # lower threshold
    t_high       = 1e-4,       # upper threshold
    use_log      = True,       # log10 scale
    high_is_green= True,       # high value = green
    mirror_x     = False,
    mirror_y     = False,
    rot_deg      = 0,          # 0 / 90 / 180 / 270
)

# 2. Optional progress callback
def on_progress(current, total, message):
    print(f"[{current}/{total}] {message}")

# 3. Run the pipeline (blocking)
pdf_path, csv_path, pptx_path = generate_report(
    filepath    = "measurements.csv",
    config      = config,
    out_dir     = "/tmp/wafer_output",
    on_progress = on_progress,
)

print(f"PDF  → {pdf_path}")
print(f"CSV  → {csv_path}")
print(f"PPTX → {pptx_path}")
```

Output directory is created if it does not exist. All three output files are written to out_dir with timestamped names.

Cancellation (headless): raise RuntimeError from the on_progress callback at any point to abort the pipeline cleanly.

10. Export Format Specifications

10.1 PDF

Property	Value
Page size	A4 (210 × 297 mm), portrait
Summary pages	~55 lines each, monospace font
Chart page	8-condition bar chart, full page
Wafer pages	5 × 5 grid = 25 wafers per page
Bookmarks	Summary Report, 8-Condition Distribution, one per lot
File naming	{basename}{log_suffix}_{timestamp}.pdf

10.2 CSV

```
Lot ID,Wafer ID,Green,Yellow,Red
LOT001,1,1,0,0
LOT001,2,1,1,1
LOT001,3,0,1,1
...
```

- One row per wafer with ≥4 die points.
- Green/Yellow/Red: 1 if ≥1 die in that zone, else 0.
- File naming: {basename}{log_suffix}_color_summary_{timestamp}.csv

10.3 PPTX

Slide	Content
1	Title slide with dataset name
2 … N	Summary slides (8-condition bar chart)
N+1 … end	Individual wafer slides (one PNG per slide, sorted A–Z)

- Template: Infineon Default Theme.pptx if found; python-pptx blank otherwise.
- File naming: {basename}{log_suffix}_{timestamp}.pptx
- Requires: py -m pip install python-pptx

11. Configuration Quick Reference

11.1 PlotConfig Fields

Field	Type	Default	Description
t_low	float	—	Lower threshold (any finite float, scientific notation OK)
t_high	float	—	Upper threshold (must differ from t_low)
use_log	bool	False	Log10 scale applied to measurement values and grid
high_is_green	bool	False	True → high=green/low=red; False → high=red/low=green
mirror_x	bool	False	Horizontally flip the wafer map
mirror_y	bool	False	Vertically flip the wafer map
rot_deg	int	0	Clockwise rotation in degrees: 0, 90, 180, or 270

11.2 Constants (config.py)

Constant	Value	Where Used
GRID_RESOLUTION	100	geometry.build_wafer_grid() — grid dimensions
WAFER_RADIUS_FACTOR	1.05	geometry.build_wafer_grid() — radius expansion
MIN_POINTS_FOR_PLOT	4	plotting.draw_wafer_ax() — skip threshold
MAX_WAFERS_PER_PAGE	25	pdf_exporter — wafers per page
LINES_PER_SUMMARY_PAGE	55	pdf_exporter — text lines per summary page

11.3 Log File Location

```
E:\claude\Wafer_tool_NEW2\wafer_tool.log
```

Append-only. A new session separator is written each time main.py starts. To read it after a crash or error, open the file in any text editor or share it for remote diagnosis.

12. Building the Executable (.exe)

The tool ships two PyInstaller build configurations. Both are Windows 64-bit and require no Python installation on the target machine. PyInstaller bundles the Python interpreter, all dependencies (PyQt6, Matplotlib, SciPy, NumPy, Pandas, ReportLab, python-pptx, pypdf), and all Matplotlib/SciPy data files into the output.

12.1 Prerequisites

- Python 3.11+ with all project dependencies installed (py -m pip install -r requirements.txt).
- PyInstaller 6.x: py -m pip install pyinstaller
- Run all build commands from the repository root (E:\claude\Wafer_tool_NEW2\).

12.2 Option A — Standalone Single-File EXE

Everything packed into one **114 MB** .exe. Copy the single file to any Windows PC and double-click — no folder, no DLLs.

Spec file: WaferMapTool.spec

```
py -m PyInstaller WaferMapTool.spec --noconfirm
```

Output:

```
dist\WaferMapTool.exe          (114 MB — single file)
```

Trade-off: On first launch the bootloader extracts all files to the OS temp directory (%TEMP%_{MEI...}). This adds ~3–8 seconds to the very first startup. Subsequent launches from the same temp extraction are faster. Antivirus software may flag single-file PyInstaller exes — add an exception if needed.

12.3 Option B — Folder Distribution (small EXE + lib folder)

A **23 MB** launcher exe sits alongside a _internal/ folder containing all DLLs and data files (~262 MB total). Faster startup than the single-file version (no extraction step).

Spec file: WaferMapTool_folder.spec

```
py -m PyInstaller WaferMapTool_folder.spec --noconfirm
```

Output:

```
dist\WaferMapTool_folder\
    WaferMapTool.exe        (23 MB — launcher)
    _internal\              (239 MB — DLLs, data, Python runtime)
        PyQt6\
        matplotlib\
        scipy\
        numpy\
        pandas\
        ... (all dependencies)
```

To distribute: Zip the entire WaferMapTool_folder\ directory. The small exe will **not** run if separated from its _internal\ folder.

12.4 Comparison

Property	Standalone EXE	Folder Distribution
File to share	1 file (114 MB)	1 folder (~262 MB zipped)
EXE size	114 MB	23 MB

Property	Standalone EXE	Folder Distribution
First launch	~5–10 s (extraction)	~2–3 s (direct load)
Subsequent launch	~2–3 s	~2–3 s
Antivirus risk	Moderate	Low
Easy to update	Rebuild whole exe	Replace individual files
Spec file	WaferMapTool.spec	WaferMapTool_folder.spec
Build command	PyInstaller WaferMapTool.spec	PyInstaller WaferMapTool_folder.spec

12.5 What Is Bundled

Package	Version	Purpose in App
Python	3.11	Runtime interpreter
PyQt6	latest	GUI framework (widgets, signals, QThread)
Matplotlib	latest	Wafer contour maps, PDF pages, histogram/scatter
NumPy	latest	Float32 array maths
SciPy	latest	griddata interpolation (Delaunay triangulation)
Pandas	latest	DataFrame for wafer data (CSV parsing)
pypdf	latest	PDF bookmark injection
python-pptx	latest	PowerPoint report generation (optional)
ReportLab	4.x	Documentation PDF generator
PyInstaller	6.x	Build tool (not included in output)

12.6 Rebuild After Code Changes

After modifying any .py file, rebuild with the same command. PyInstaller detects changed files automatically:

```
cd E:\claude\Wafer_tool_NEW2

# Rebuild standalone (replaces dist\WaferMapTool.exe)
py -m PyInstaller WaferMapTool.spec --noconfirm

# Rebuild folder version (replaces dist\WaferMapTool_folder\)
py -m PyInstaller WaferMapTool_folder.spec --noconfirm
```

The build\ cache is reused between rebuilds to speed up the process. Delete build\ manually if you encounter stale-cache issues.

12.7 Adding a Custom Icon

To set a custom taskbar/exe icon, supply a .ico file and uncomment the icon line in both spec files:

```
# In WaferMapTool.spec or WaferMapTool_folder.spec, inside EXE():
icon="wafer_tool.ico",    # path relative to repo root
```

Convert a PNG to ICO online (e.g., convertio.co) or with Pillow:

```
from PIL import Image
img = Image.open("wafer_tool.png")
```

```
img.save("wafer_tool.ico", format="ICO",
         sizes=[(16,16),(32,32),(48,48),(256,256)])
```